WORKED SOLUTION

STEP III 2025, Question 10

Mechanics • Friction on Moving Plank

STEP III 2025, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Solve the differential equation for h first: it is simple harmonic motion, and it says that the plank's angle of inclination grows at the steady rate ω. Once the angle is known, the particle's position is a fixed fraction of the way along the plank, so differentiating twice gives its acceleration. The forces then follow from Newton's second law, resolved perpendicular to and along the plank, which is much tidier than resolving horizontally and vertically.

Solution

We model the plank as a rigid rod and P as a particle. Let θ be the angle the plank makes with the horizontal.

Part (i)

The equation $\ddot{\mathrm{h}} = -\omega^2 \mathrm{h}$ has general solution $\mathrm{h} = a\cos\omega t + b\sin\omega t$. The condition $\mathrm{h}(0) = 0$ gives $a = 0$, and $\dot{\mathrm{h}}(0) = b\omega = \omega L$ gives $b = L$, so

$$\mathrm{h}(t) = L\sin\omega t.$$

Since A is at height h and the plank has length L, $\sin\theta = \mathrm{h}/L = \sin\omega t$, so $\theta = \omega t$ (for $0 \le \omega t \le \frac{\pi}{2}$, which is the whole of the motion: after that A would have to rise above height L). Thus $A = (0, L\sin\omega t)$ and $B = (L\cos\omega t, 0)$.

The point C is the fraction s of the way from A to B, so $C = A + s(B - A)$:

$$x = sL\cos\omega t, \qquad y = (1-s)L\sin\omega t.$$

The first of these is the required x-coordinate, and the second is the matching expression for the y-coordinate.

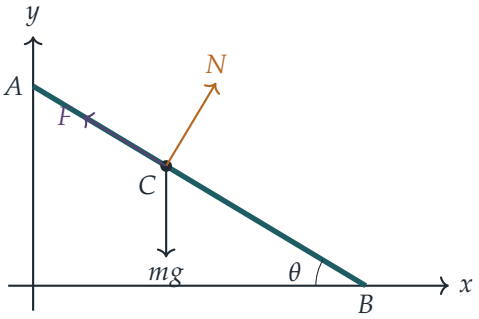
Part (ii)

Differentiating twice with respect to t,

$$\ddot{x} = -sL\omega^2 \cos\omega t, \qquad \ddot{y} = -(1-s)L\omega^2 \sin\omega t.$$

Part (iii)

The forces on P are its weight mg downwards, the normal reaction N perpendicular to the plank (upwards), and friction F along the plank, which we take as positive when it points up the slope, towards A.



The unit vector perpendicular to the plank (upwards) is $(\sin\theta, \cos\theta)$, and the unit vector along the plank towards A is $(-\cos\theta, \sin\theta)$, with $\theta = \omega t$. Resolving the equation of motion $m(\ddot{x}, \ddot{y}) = N(\sin\theta, \cos\theta) + F(-\cos\theta, \sin\theta) + (0, -mg)$ in these two directions (that is, taking the dot product with each unit vector):

Perpendicular to the plank.

$$\begin{aligned} N - mg\cos\theta &= m(\ddot{x}\sin\theta + \ddot{y}\cos\theta) \\ &= -mL\omega^2(s + (1-s))\sin\theta\cos\theta = -mL\omega^2 \sin\theta\cos\theta. \end{aligned}$$

With $L\omega^2 = kg$, this gives

$$N = mg\cos\theta - mgk\sin\theta\cos\theta = mg(1 - k\sin\omega t)\cos\omega t,$$

as required.

Along the plank, towards A.

$$\begin{aligned} F - mg\sin\theta &= m(-\ddot{x}\cos\theta + \ddot{y}\sin\theta) \\ &= mL\omega^2(s\cos^2\theta - (1-s)\sin^2\theta) = mgk(s - \sin^2\theta), \end{aligned}$$

using $\cos^2\theta + \sin^2\theta = 1$. Hence

$$F = mgsk + mg\sin\theta(1 - k\sin\theta) = mgsk + \underbrace{mg(1 - k\sin\theta)\cos\theta}_{N}\tan\theta,$$

that is, $F = mgsk + N\tan\omega t$, as required. Notice that F comes out positive whenever $N > 0$, so friction always acts up the plank while P is pressed against it.

Part (iv)

Write $\mu = \tan\alpha$. While $N > 0$, the particle stays at C without slipping exactly when $|F| \le \mu N$; since $F > 0$ there, this is $F/N \le \tan\alpha$. Dividing the result of part (iii) by N,

$$\frac{F}{N} = \frac{sk}{(1 - k\sin\theta)\cos\theta} + \tan\theta.$$

Initially, $\theta = 0$, so $N = mg > 0$ and $F/N = sk$. So the particle does not slip at the start provided $sk < \tan\alpha$.

Ask yourself Why must the ratio F/N eventually reach $\tan\alpha$ before N vanishes?

Now suppose $sk < \tan\alpha$. The factor $\cos\theta$ vanishes at $\theta = \frac{\pi}{2}$, and if $k > 1$ the factor $1 - k\sin\theta$ vanishes earlier, at $\sin\theta = 1/k$. Let θ_0 be the first angle at which $N = 0$ (either $\frac{\pi}{2}$ or $\arcsin(1/k)$). On $0 \le \theta < \theta_0$ we have $N > 0$, and F/N is continuous. As $\theta \to \theta_0^-$, the denominator $(1 - k\sin\theta)\cos\theta$ tends to 0 through positive values while the numerator sk is a fixed positive number, so the first term tends to $+\infty$ (and $\tan\theta$ is either bounded or also tends to $+\infty$). Hence $F/N \to +\infty$.

So F/N starts at $sk < \tan\alpha$ and grows without bound before N reaches zero. By the intermediate value theorem there is a first angle $\theta_1 < \theta_0$ at which $F/N = \tan\alpha$, and the particle begins to slip there. This happens at an angle where N is still positive, which is the first claim.

At this angle, since $sk > 0$ and $(1 - k\sin\theta_1)\cos\theta_1 > 0$,

$$\tan\alpha = \frac{F}{N} = \frac{sk}{(1 - k\sin\theta_1)\cos\theta_1} + \tan\theta_1 > \tan\theta_1.$$

Both α and θ_1 lie in $[0, \frac{\pi}{2})$, where tan is increasing, so $\theta_1 < \alpha$: the plank makes an angle less than α with the horizontal when slipping starts. This is the second claim.

The second claim is less surprising than it looks. A particle at rest on a fixed plank slips exactly when the plank's angle reaches α, the angle of friction. Here the plank is accelerating, and the extra force needed to carry the particle along with it (the $mgsk$ term) means friction gives out sooner.

Checks

- At $t = 0$ the plank is horizontal and the particle's acceleration is $(-sL\omega^2, 0)$, horizontal. Then $N = mg$ and $F = msL\omega^2 = mgsk$ should provide exactly that acceleration, and they do.
- If $\omega \to 0$ (so $k \to 0$), the plank is raised very slowly; then $N \to mg\cos\theta$ and $F \to mg\sin\theta$, the familiar equilibrium values on a fixed slope, and slipping happens at $\theta = \alpha$, the boundary of our inequality.
- Numerically, with $s = \frac{1}{2}$, $k = \frac{1}{2}$ and $\alpha = \frac{\pi}{4}$ (so $sk = \frac{1}{4} < 1$), solving $F/N = 1$ shows slipping at $\theta \approx 0.545$ rad, about 31°, where $N \approx 0.63mg > 0$ and $\theta < 45^\circ$.

A computer check over thousands of random choices of s, k and α with $sk < \tan\alpha$ found slipping at an angle below α with $N > 0$ every time.

Common slips

- Taking θ to be the angle at A instead of at B: then the sines and cosines swap, and the given expressions will not come out.
- Resolving horizontally and vertically and then trying to solve two simultaneous equations with $\sin\theta$ and $\cos\theta$ everywhere. Resolving along and perpendicular to the plank separates N and F immediately.
- In part (iv), arguing only from θ approaching $\frac{\pi}{2}$ and forgetting that when $k > 1$ the normal reaction vanishes earlier, at $\sin\theta = 1/k$.

Beyond STEP

This question is about forces in a moving frame: from the plank's point of view, the particle feels an extra "fictitious" force equal to $-m$ times the acceleration of the point C, and the term $mgsk$ is exactly that inertial force at the start. At university, this idea is developed into the full theory of accelerating and rotating reference frames, including the centrifugal and Coriolis forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames with many worked examples.

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